Power BI Data Analytics Report: Business Performance Insights for FinServe

A Group Project for BAA1026 on Data Visualisation and Analysis

Submitted To:

Babu Veeresh Thummadi

Dublin City University - Dcu Business School

Module:

BAA1026 - Data Analytics: Programming and Visualisation

Group Number:

Group 6

Group Members:

Aaron Hendrick - 22453882

David Odekunle - 22386001

Dennis Chumakov - 46511

Friedrich Zürger - 46544

Robert Mullins - 22352273

Submission Date:

11 November 2025

Word Count Excluding Appendices & References:

1,955

Executive Summary

This analytical report evaluates the financial and operational performance of FinServe from 2015 to 2024. The goal is to inform strategic decisions on regional expansion and product management by examining revenue, costs, and regional trends. The analysis is based on the *powerbi_data_group6_FinServe_Group.xlsx* data set, which was cleaned and standardised to remove inconsistencies and ensure the reliability of calculated metrics.

Key findings indicate a steady revenue increase from 2015-2022, slightly dropping in 2020 before bouncing back. On the other hand, profitability peaked in 2020 before declining due to rising costs and excessive discounting. Regional analysis shows that Africa generated the highest revenue but had weak profitability, suggesting operational inefficiencies. In contrast, Europe and North America achieved higher profit margins. Volatility was also observed in certain product categories, notably construction, energy, and travel.

Recommendations include focusing on high-margin regions such as Europe and North America, reducing excessive discounting, pursuing growth opportunities in select emerging markets, and improving data collection to better monitor product volatility. These measures aim to enhance FinServe's long-term profitability and stability.

Table of Contents

Executive Summary 2
1 - Introduction 4
2 - Data Preparation and Methodology 4
3 - Dashboard Design and Visualisation Approach 5
4 - Insights and Analysis 6
4.1 Overall Performance 6
4.2 Regional Analysis 7
4.3 Product and Category Analysis 8
4.4 Trend and Pattern Analysis 10
5 - Recommendations 11
6 - Conclusion 11
References 13
Appendix A – FinServe Dashboard 14

1 - Introduction

The purpose of this Power BI project is to analyse FinServe's financial and operational performance over the ten years from 2015 to 2024. The analysis provides a data-driven understanding of revenue, profit, and regional trends while identifying key drivers influencing profitability. This report serves as both a diagnostic and strategic tool to guide decisions on regional expansion, cost optimisation, and product portfolio management.

The dataset, *powerbi_data_group6_FinServe_Group.xlsx*, includes transactional and performance-related data such as revenue, profit, expenses, discounts, product categories, and regional metrics. Each observation represents combined financial data across years and regions, enabling a comprehensive exploration of FinServe's business performance.

Power BI was selected for its strong data modelling, visual analytics, and interactivity capabilities. Data preprocessing and validation steps were completed using Microsoft Excel and Power Query before building the dashboard in Power BI. This ensured accuracy, consistency, and scalability.

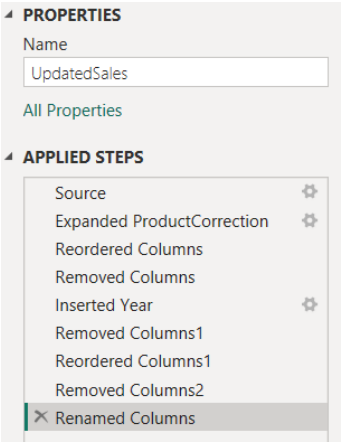
The report details data preparation and methodology in Section 2; dashboard design and visualisation in Section 3; analytical insights in Section 4; and strategic recommendations in Section 5. The report concludes with key takeaways on FinServe's business performance and opportunities. In the appendices, you will find supporting visuals.

2 - Data Preparation and Methodology

The data preparation process combined both technical transformation and exploratory analysis to ensure the dataset was accurate, consistent, and ready for visualisation. The approach was guided by *Exploratory Data Analysis (EDA)* principles (IBM, 2021) and Microsoft's *Power Query Editor* framework (Microsoft, n.d.), which together define both the rationale and process of transformation.

The dataset was loaded into Power BI and accessed through Transform Data, which opens the Power Query Editor. According to Microsoft (n.d.), this feature connects to data sources and allows shaping and transformation before modelling. An initial EDA identified inconsistencies – misaligned Country and Region fields and incorrect Product-Category pairings.

To correct these issues, verification tables were created in Excel and merged into Power BI using a Left Join, ensuring all original records were retained while replacing inaccurate values. Irrelevant columns were dropped, the Date field was converted to Year, and unnecessary fields such as Customer Name, Customer ID, and Order Priority were removed.



Product	CorrectCategory
Building Material	Construction
Car	Automotive
Flight	Travel
Ticket	Travel
Insurance Plan	Finance
Monitor	Electronics
Laptop	Electronics
Printer	Electronics
Smartphone	Electronics
Solar Panel	Energy
Tablet	Electronics

Country	CorrectRegion
Brazil	South America
Germany	Europe
USA	North America
South Africa	Africa
India	Asia

Further inconsistencies were identified in key financial metrics such as Total Revenue, Total Cost, Total Profit, and Net Profit. Because the dataset did not have formulas and the calculations displayed irregular logic, standardised measures were defined as follows:

- Standard Revenue = Units Sold x Unit Price
- Standard Total Cost = (Units Sold x Unit Cost) + (Units Sold x Shipping Cost)
- Standard Total Profit = Standard Revenue - (Units Sold x Unit Cost)
- Standard Net Profit = Standard Revenue - Standard Total Cost

These measures were implemented as calculated columns in Power BI. Every step was recorded in the Applied Steps pane, ensuring a transparent and reproducible workflow. Integrating EDA with Power BI's transformation tools established a strong foundation for accurate and meaningful analysis.

3 - Dashboard Design and Visualisation Approach

The Power BI dashboard was designed to provide an interactive, structured and insightful overview of FinServe's performance from 2015 to 2024. It consists of four

interconnected pages (which can be found in Appendix A) – Executive Overview, Regional & Market Insights, Product & Category Analysis, and Trend/Pattern Analysis – each focusing on a specific analytical dimension.

Interactive slicers for year, region, and category enhance usability and consistency across pages.

- Executive Overview: Displays key performance indicators (KPIs) for Total Revenue, Total Cost, Total Profit, Net Profit, and Profit Margin. Supporting visuals include line and column charts showing trends in profit, revenue, and cost, and a bar chart comparing revenue by region.
- Regional & Market Insights: Features clustered column charts comparing revenue, cost, and profit across regions, complemented by a line chart for profit margin trends. A donut chart and decomposition tree highlight each region's contribution to overall profitability.
- Product & Category Analysis: Uses bar and donut charts to visualise profitability by category and a scatter plot to reveal relationships between sales volume and profitability.
- Trend & Pattern Analysis: Combines column, line, and combo charts to assess year-over-year performance stability, supported by KPIs for average unit, cost, and shipping values.

This consistent design ensures a logical progression from high-level performance metrics to granular insights.

4 - Insights and Analysis

4.1 Overall Performance

Between 2015 and 2024, FinServe generated a total revenue of €126.65 million and incurred total expenses of €109.3 million, resulting in a net profit of €17.34 million and an overall profit margin of 13.69%. While revenue displayed steady growth, operational costs increased at a faster rate, leading to margin compression. Profitability peaked in 2020 before declining in subsequent years due to rising costs and extensive discounting practices.

Year-over-year analysis highlights this pattern clearly. The years leading up to 2020 showed consistent profit expansion, reflecting a period of strong demand and pricing

stability. However, post-2020, the company faced increasing cost pressures, particularly in shipping and production, which limited profit growth despite continued revenue gains.

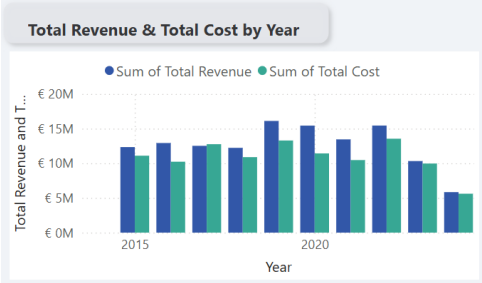


Figure 1: Year-over-Year Revenue and Cost Growth (2015-2024)

Regional profit margins fluctuated between 10% and 20% over the decade. Europe and North America recorded the strongest margins at 19.28% and 19.78%, respectively, suggesting operational efficiency and effective pricing. Meanwhile, Africa generated the highest total revenue but suffered from disproportionately high costs, reducing profitability. These findings indicate that FinServe's growth in sales volume has not consistently translated into profit growth, emphasising the importance of managing operational costs and regional efficiency to sustain long-term profitability.

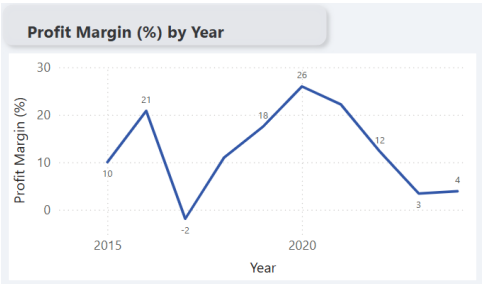


Figure 2: Profit Margin Fluctuation and 2020 Profit Peak

4.2 Regional Analysis

Across all product categories from 2015 to 2024, FinServe's regional performance varied significantly. Africa consistently achieved the highest revenue and units sold among all regions, yet profitability remained relatively low. This pattern suggests smaller margins or higher operating costs compared to other markets. In contrast, Europe and North America achieved the highest overall profits - €6.16M and €6.18M, respectively – despite lower sales volumes, indicating stronger pricing power and greater cost control.

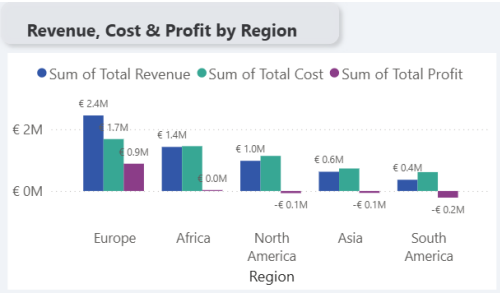


Figure 3: Regional Comparison of Revenue, Cost and Profit (2024)

Asia performed robustly across most categories, doing particularly well in Energy, delivering 57.42% of the company's total profits in that sector. Europe stood out in high-value sectors such as Electronics and Finance, contributing 32.29% and 30.15% of total profits, respectively. North America performed similarly well, excelling in Travel (44.69%) and Finance (26.02%). By contrast, South America underperformed, with several categories recording losses.

These findings highlight how differences in market maturity, cost structures, and product composition influence profitability across regions. For example, Construction in Africa is characterised by high-volume, low-margin operations, while in Europe, the same category achieves profitability through lower volumes and higher margins. This underscored the need for region-specific pricing and cost management to optimise overall returns.

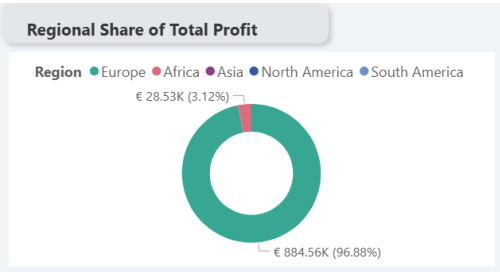


Figure 4: Regional Share of Total Profit (2024)

4.3 Product and Category Analysis

From 2015 to 2024, Electronics emerged as FinServe's most profitable product category, accounting for 37% of total profits over the decade and 72% of total profits in 2024, primarily driven by strong European performance. However, this heavy reliance on a single category and region poses a potential concentration risk.

Figure 5: Revenue and Profit Distribution by Category (2015-2024)

Insurance Plans have consistently been the most purchased individual product by volume, though sales have declined slightly in recent years. A more concerning trend is the extreme volatility in product profitability. For example, Construction Materials generated over 35% of profits in 2023 but fell to just 0.25% in 2024. Similarly, Energy and Travel products, which contributed 35% and 29% of profits respectively, both declined to near zero in 2024. These abrupt changes suggest weaknesses in cost control and potentially volatile input costs.

Figure 6: Energy Sales and Profit Relationship Energy (2022-2024)

Figure 7: Travel Sales and Profit Relationship (2022-2024)

The Automotive category also demonstrated inconsistent profitability across the years. The Travel category is particularly problematic, having generated minimal or negative profits since 2022. Given this sustained underperformance, FinServe should consider discontinuing or restructuring this product line to minimise losses. Instead, the company should focus on optimising cost management and strengthening performance in historically profitable categories such as Electronics and Automotive.

Figure 8: Electronics Sales and Profit Relationship (2022-2024)

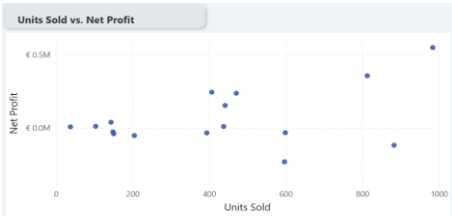


Figure 9: Automotive Sales and Profit Relationship (2022-2024)

4.4 Trend and Pattern Analysis

A temporal analysis of FinServe's performance between 2015 and 2024 reveals distinct performance cycles and emerging patterns. Revenue increased steadily until 2020, when profitability reached its peak, before declining due to escalating operating costs. This suggests a post-2020 stabilisation phase, in which cost management became a primary driver of performance.

Figure 10: Revenue and Profit Margin Trends (2015-2024)

Profit margins fluctuated between 10% and 20%, reflecting varying efficiency levels in cost control. Seasonal variations were evident, with higher revenue typically occurring mid-year, likely linked to cyclical demand in consumer and technology sectors.

A key finding is the inverse relationship between discount rates and profit margins. Aggressive discounting led to temporary boosts in sales but ultimately eroded profitability. As illustrated in Figure 11, revenue grew from 2020 – 2022, but profit margins spiked downwards because of excessive discounting and cost escalation. These findings reinforce the need for balanced pricing strategies that align with operational efficiency to sustain long-term profitability.

Figure 11: Revenue and Profit Margin Trends (2020-2024)

5 - Recommendations

1. *Focus on High-Margin Regions*

FinServe should prioritise investment in Europe and North America, which have demonstrated strong profitability despite lower sales volumes. Concentrating resources in these regions will foster sustainable growth and reduce exposure to high-cost, low-margin markets (Teece, 2022).

2. *Refine Pricing Strategy*

The company should shift away from broad discounting - particularly in low-margin regions such as Africa – and adopt a value-based pricing approach. Studies show that value-based pricing improves both profitability and long-term financial performance compared to cost-plus or competition-based methods (De Toni et al., 2017). Implementing a structured, data-informed pricing model will help FinServe mitigate the negative effects of discounting on margins.

3. *Pursue Selective Expansion*

FinServe should consider staged expansion into emerging markets where specific categories perform well, such as Energy in Asia. This cautious, data-driven approach minimises strategic risk and enables the company to evaluate product-market fit before committing significant resources (Benito et al., 2022).

4. *Improve Data Quality and Analytics Capabilities*

FinServe should invest in better data collection and analytical tools to track product volatility and regional performance more precisely. Enhanced data quality enables improved forecasting, greater operational efficiency, and more informed strategic decision-making – all of which are essential for managing fluctuating profit margins (Wang et al., 2023).

6 - Conclusion

The Power BI dashboard provided a comprehensive view of FinServe's performance from 2015 to 2024, revealing key trends across regions, product categories, and time periods. The analysis highlighted Africa's strong sales but limited profitability, Europe and North America's operational efficiency, and Asia's dominance in the Energy sector.

Profit margins fluctuated due to rising costs and inconsistent product performance, particularly after 2020. However, the dashboard's interactivity offered valuable insights into the complex relationships between revenue, cost, and profit, enhancing understanding of FinServe's operational dynamics.

Looking ahead, FinServe should continue leveraging Power BI to deepen its analytical capabilities. Incorporating predictive forecasting, customer segmentation, and cost modelling could provide forward-looking insights to anticipate market changes and support data-driven strategies that strengthen long-term financial performance and stability.

References

Benito, G. R. G., Cazurra, A. C., Mudambi, R., Pedersen, T., & Tallman, S. (2022). *The Future of Global Strategy.* Global Strategy Journal, 12(3), 421–450. Wiley. Available at: https://sms.onlinelibrary.wiley.com/doi/10.1002/gsj.1464

Query Overview in Power BI Desktop—Power BI | Microsoft. (n.d.). Retrieved November 8, 2025, from https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-query-overview

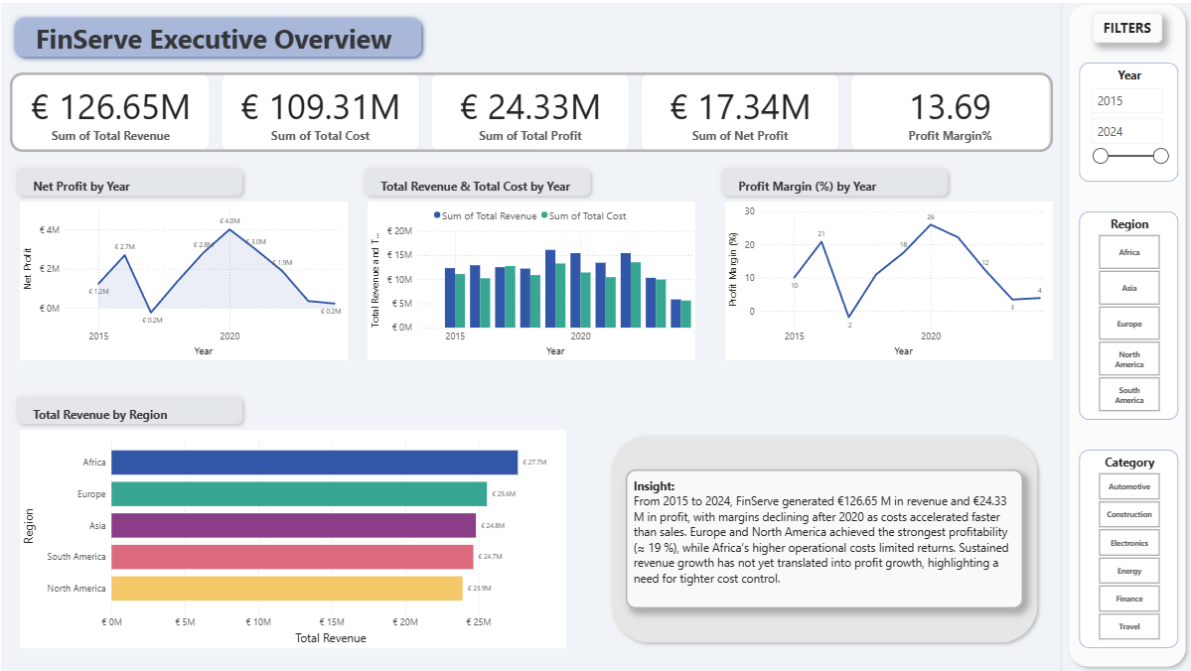
Teece, D.J. (2022*). A wider-aperture lens for global strategic management: The multinational enterprise in a bifurcated global economy.* Global Strategy Journal, 12(3). doi:https://doi.org/10.1002/gsj.1462.

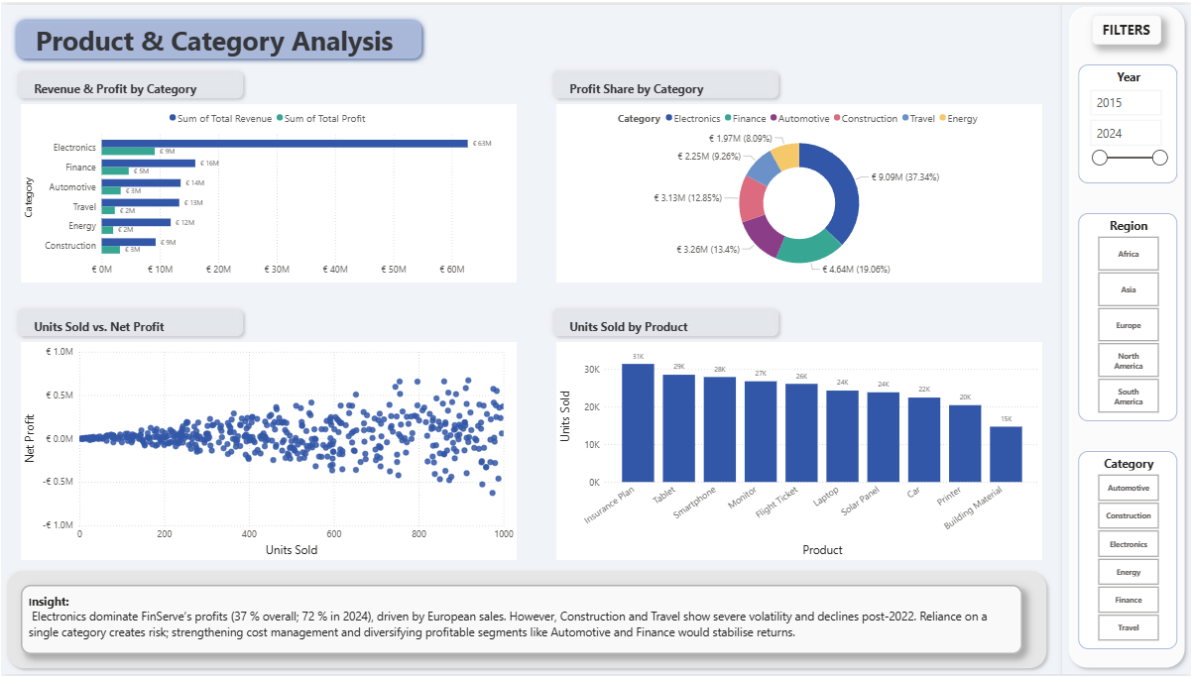
Toni, D.D., Milan, G.S., Saciloto, E.B. and Larentis, F. (2017). *Pricing Strategies and Levels and Their Impact on Corporate Profitability*. Revista de Administração, [online] 52(2), pp.120–133. doi:https://doi.org/10.1016/j.rausp.2016.12.004.

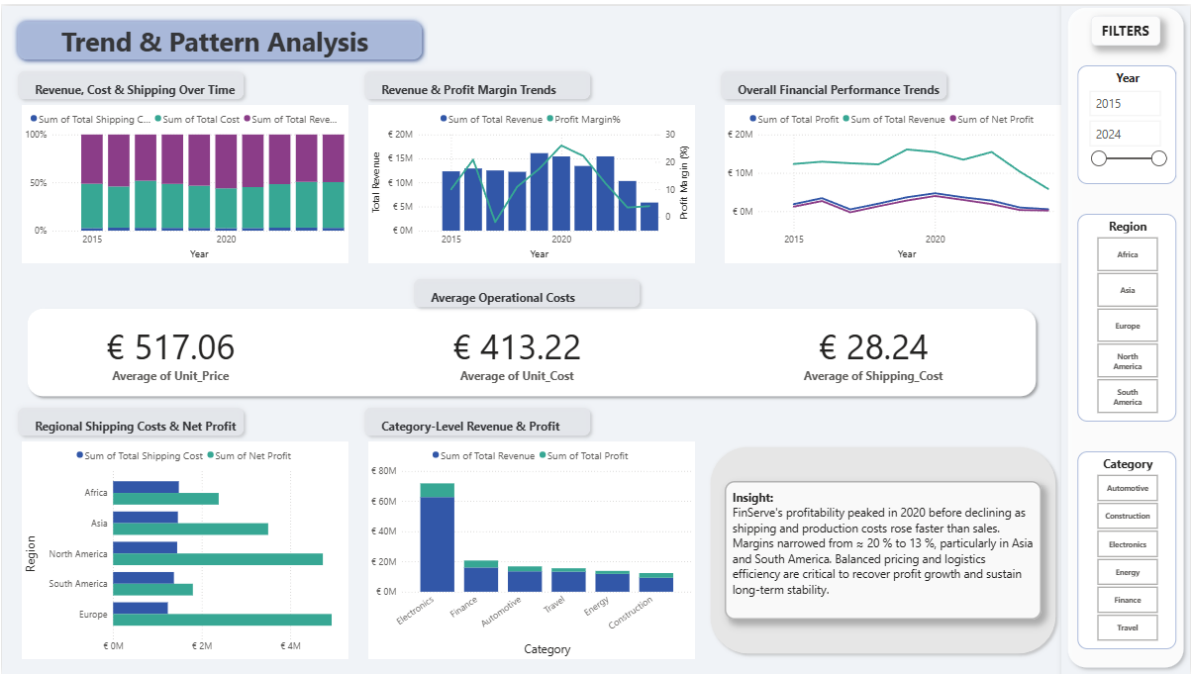
Wang, J., Liu, Y., Li, P., Lin, Z., Sindakis, S. and Aggarwal, S. (2023). *Overview of Data Quality: Examining the Dimensions, Antecedents, and Impacts of Data Quality.* Journal of the Knowledge Economy, [online] 15. Available at: https://link.springer.com/article/10.1007/s13132-022-01096-6.

What is Exploratory Data Analysis? | IBM. (2021, October 4). https://www.ibm.com/think/topics/exploratory-data-analysis

Appendix A – FinServe Dashboard




Product & Category Analysis
Revenue & Profit by Category
Sum of Total Revenue
Sum of Total Profit
Category
Electronics
Finance
Automotive
Travel
Energy
Construction
€ 0M
€ 10M
€ 20M
€ 30M
€ 40M
€ 50M
€ 60M
Profit Share by Category
Category
Electronics
Finance
Automotive
Construction
Travel
Energy
€ 1.97M (8.09%)
€ 2.25M (9.26%)
€ 9.09M (37.34%)
€ 3.13M (12.85%)
€ 3.26M (13.4%)
€ 4.64M (19.06%)
Units Sold vs. Net Profit
Net Profit
Units Sold
Units Sold by Product
Units Sold
Product
Insurance Plan
Tablet
Smartphone
Monitor
Flight Ticket
Laptop
Solar Panel
Car
Printer
Building Material
Insight:
Electronics dominate FinServe's profits (37 % overall; 72 % in 2024), driven by European sales. However, Construction and Travel show severe volatility and declines post-2022. Reliance on a single category creates risk; strengthening cost management and diversifying profitable segments like Automotive and Finance would stabilise returns.
FILTERS
Year
2015
2024
Region
Africa
Asia
Europe
North America
South America
Category
Automotive
Construction
Electronics
Energy
Finance
Travel


Trend & Pattern Analysis
Revenue, Cost & Shipping Over Time
Sum of Total Shipping C...
Sum of Total Cost
Sum of Total Reve...
Year
Revenue & Profit Margin Trends
Sum of Total Revenue
Profit Margin%
Total Revenue
Profit Margin (%)
Year
Overall Financial Performance Trends
Sum of Total Profit
Sum of Total Revenue
Sum of Net Profit
Year
Average Operational Costs
€ 517.06
Average of Unit_Price
€ 413.22
Average of Unit_Cost
€ 28.24
Average of Shipping_Cost
Regional Shipping Costs & Net Profit
Sum of Total Shipping Cost
Sum of Net Profit
Region
Africa
Asia
North America
South America
Europe
Category-Level Revenue & Profit
Sum of Total Revenue
Sum of Total Profit
Category
Electronics
Finance
Automotive
Travel
Energy
Construction
Insight:
FinServe's profitability peaked in 2020 before declining as shipping and production costs rose faster than sales. Margins narrowed from ≈ 20 % to 13 %, particularly in Asia and South America. Balanced pricing and logistics efficiency are critical to recover profit growth and sustain long-term stability.
FILTERS
Year
2015
2024
Region
Africa
Asia
Europe
North America
South America
Category
Automotive
Construction
Electronics
Energy
Finance
Travel